

Mohamad Khawam

New Jersey, United States | ✉ Khawammohamad99@gmail.com | ☎ (862)-285-1846 | 🌐 github.com/mkhawam | 🔗 linkedin.com/in/mohamad-k | 🌐 mohamadk.com

Summary

Backend and platform engineer, and the sole active maintainer of a Django/DRF grading platform with 1,950+ users. I own everything from its sandboxed multi-language execution engine to the layer that tracks AI spend per token. Most of my work lives in concurrency, query performance, container isolation, and application security.

Experience

Rutgers University, Application Developer (Software Engineer)

June 2025 - Present

- Sole active engineer on codePost, a Django/DRF grading platform serving 1,950+ students, graders, and course admins across 19 courses. The backend is 80K lines of Python running as 12 containers across 4 VMs.
- Built the engine that runs untrusted student code: one Docker container per run across seven language runtimes and Jupyter notebooks, with no network, all capabilities dropped, and hard caps on CPU, memory, PIDs, and wall time.
- Cut redundant grading compute: unchanged submissions skip the container run entirely via a content-hash cache, and test suites are batched so each category script runs once instead of once per test case.
- Fixed N+1 queries in the permission chain with per-request role memoization and eager loading at 77 call sites, and made concurrent AI jobs idempotent.
- Built per-token cost tracking for the platform's AI features: provider rate tables with cached-token discounts, fixed-point charges quantized to a millionth of a dollar, per-course rate overrides, and hourly, daily, and monthly rollups.
- Made autograder environments self-healing, promoting images above an 0.8 success rate and rolling back below 0.5, and wrote the five-probe health endpoint that drives container autoheal and staged deploys. Grew the backend suite to 992 tests behind a four-job CI gate.
- Sole maintainer of the platform's 103K-line legacy React SPA. Migrated it from Create React App to Vite and React Router v5 to v7, added lazy-loaded route splitting, and gated deploys on a strict type check plus 173 tests.
- Contributed 337 commits to the university's shared Ansible fleet repository, rebuilding the JupyterHub role with pluggable Kerberos, CAS, and Azure AD auth and per-user systemd cgroups that cap memory, disk, and CPU time.
- Wrote and published jupyter-assignments to npm and PyPI, a JupyterLab extension that fronts three grading backends behind one adapter interface and enforces a four-tier role hierarchy, covered by 320 tests.
- Fixed a directory traversal in dataset uploads, an email-and-logging infinite recursion, and authorization gaps that let students run code and see draft grades.

Rutgers University, Student Lab Technician

January 2023 - May 2025

- Ran 10 Hackerspace workspaces, writing student-facing documentation and maintaining the WordPress site as part of modernizing the space.
- Built a Raspberry Pi computer-vision learning station with MediaPipe, used for walk-in student workshops.
- Built a 3D printer monitoring service (Express and React) streaming live video of the printer bay so students could check job status without walking over.

Swish, Full-Stack Developer

July 2021 - December 2021

- Automated account creation on web platforms with Puppeteer and Node.js, with Mocha regression tests covering the signup flows.
- Migrated the CLI to an Electron desktop application, replacing flag-based invocation with a GUI workflow.

C-Tech, Computer Technician

July 2019 - December 2021

- Automated computer initialization with Powershell scripts, cutting build time from 40 to 20 minutes.
- Implemented a secure HDD erasure workflow processing dozens of drives daily, onboarding 3 new enterprise clients.

Skills

Languages: Python, TypeScript, JavaScript, C, Java, Golang, SQL, Bash, PowerShell

Frameworks: Django, Django REST Framework, Django Channels, Celery, Flask, Node.js, Express, React, Next.js, JupyterLab, Electron

Data: MariaDB / InnoDB, MongoDB, Redis, Django ORM, Schema Design and Migrations, Query Optimization and Indexing, Caching

Infrastructure: Docker, Docker Compose, Ansible, Nginx, Linux, systemd, Proxmox, VMware, OpenStack, AWS, GitHub Actions, GitLab CI, CI/CD, Zabbix

Security: Application Security, OWASP Top 10, RBAC and Authorization, CAS / SAML / Kerberos SSO, JWT and API Key Management, Suricata, pfSense, Wazuh

Testing: pytest, Vitest, Jest, Playwright, Mocha, axe-core, Molecule, WCAG 2.1 AA

AI: OpenAI and Gemini APIs, Ollama, LLM Cost Metering, Tool-Calling Agents

Projects

CompLock

November 2023 - Present

- Built SSH command-and-control software for networked computers in TypeScript on LevelDB and SSH2, used by RUSEC in CCDC competitions.
- Cut password rotation across 30 machines from 5 minutes to 30 seconds, with Mocha tests running in GitHub Actions.

mohamadk.com

April 2025 - Present

- Designed and built a Next.js 16 / React 19 portfolio and blog with a markdown publishing pipeline, PWA support, and a keyboard command palette.
- Built an in-browser code playground on WebContainers that boots Node.js client-side to run my published ts-declaration-json package live.
- Wrote a simulated Linux shell in TypeScript (filesystem, users, coreutils) that serves as the site's 404 page, plus an AI task-planning agent with tool calling.

AppTracker

2025

- Built a TypeScript app that tracks job applications by watching email over IMAP, classifying messages with a Bayesian classifier, and generating status updates with Ollama.
- Sends real-time status updates through a Discord bot and keeps application history in MongoDB.

Education

Bachelor of Arts, Computer Science

May 2025

- Rutgers University — Coursework: Computer Security, Software Methodology, Computer Architecture.

Leadership

RUSEC (Student Cybersecurity Club), Vice President

January 2023 - May 2025

- Directed network administration for CCDC, with the team placing 4th in 2024.
- Led club infrastructure setup on Proxmox and VMware ESXi, standing up the practice range the team trained on.
- Led development of the club's BlackBox machine environment with Wazuh detection rules and the ToolBox hardening toolkit used for CCDC preparation.